

G9: timm 1.0.29 | gdown 5.2.2, Mounted at /content/drive

SYNTHETIC=False

DATA_DIR=/content/drive/MyDrive/convergence_experiment exists=True

WORK=/content/g9_work

OUT=/content/drive/MyDrive/convergence_experiment/G9_fractal

,

PASS engine == scipy [shared] 0.711111 vs 0.711111

PASS engine == scipy [collapsed] 0.528727 vs 0.528727

PASS tie path == scipy average ranks

PASS rotation invariance

PASS shuffle null ~ 0 -0.0039

PASS R narrow bracket -> NaN

PASS R midpoint = 0.5

PASS RankBank == direct

ALL SELF-TESTS PASSED

,real run - skipped,

encoder	file key	rows_in_file	dim	norm_published	norm_file
norm_prefix identity_ok					
img_small	e1_img_ckpt_dinov2-small_cls+patch.npz	img	9533	768	54.70
54.696	54.696	True			
img_base	e1_img_ckpt_dinov2-base_cls+patch.npz	img	18066	1536	52.91
52.889	52.912	True			

img_large	e1_img_ckpt_dinov2-large_cls+patch.npz	img	23755	2048	50.83	
50.810	50.827	True				
convnext	e1_img_ckpt_convnext-base-224-22k_native.npz	img	9533	1024	16.70	
16.739	16.739	True				
bge	crossmodal_pairs.npz	txt	18066	1024	0.89	0.894
0.894	True					
gpt2	crossmodal_pairs_gpt2.npz	txt	18066	768	207.78	208.211
208.209	True					
bert	e13_txt_bert.npz	txt	18066	768	9.15	9.183
True						9.186
sbert	e13_txt_sbert.npz	txt	18066	768	2.43	2.427
True						2.428

, ranked img_small ties=0.0e+00 (15s)

ranked img_base ties=0.0e+00 (30s)

ranked img_large ties=0.0e+00 (45s)

ranked convnext ties=0.0e+00 (59s)

ranked bge ties=2.2e-08 (73s)

ranked gpt2 ties=1.3e-06 (101s)

ranked bert ties=0.0e+00 (115s)

ranked sbert ties=2.2e-08 (129s)

REPRODUCTION GATE (published E.12 / Table C12m values, tol 0.01)

pair	published	measured	delta	ok
convnext~img_small	0.397	0.3971	0.0001	True
convnext~img_base	0.392	0.3915	-0.0005	True
convnext~img_large	0.391	0.3914	0.0004	True

convnext~sbert	0.330	0.3303	0.0003	True
convnext~bert	0.305	0.3050	-0.0000	True
convnext~bge	0.302	0.3025	0.0005	True
convnext~gpt2	0.166	0.1662	0.0002	True
bge~sbert	0.768	0.7682	0.0002	True

cross-modal mean over the 16 image-text pairs: 0.290 (published 8-encoder baseline 0.290)

VERIFIED -- the pipeline reproduces the report, 9533 COCO ids, 9 ... 46492 (published range 9 ... 46492), 1000/9533

2000/9533

3000/9533

4000/9533

5000/9533

6000/9533

7000/9533

8000/9533

9000/9533

download: {'train2017': 9533} (369s)

cropped 2000/9533

cropped 4000/9533

cropped 6000/9533

cropped 8000/9533

crops: (9533, 224, 224, 3) uint8 (383s total), checkpoints found:

w_vit/deitt16_224_fractal1k_lr3e-4_300ep.pth 69 MB

w_vit/deitt16_224_fractal10k_lr3e-4_100ep.pth 90 MB

w_resnet/FractalDB-10000_res18.pth 65 MB

w_resnet/FractalDB-10000_res152.pth 315 MB

w_resnet/FractalDB-10000_dense161.pth 195 MB

w_resnet/FractalDB-10000_res50.pth 176 MB

w_resnet/FractalDB-10000_bn_alexnet.pth 397 MB

w_resnet/FractalDB-10000_res34.pth 106 MB

w_resnet/FractalDB-10000_resnext101.pth 251 MB

w_resnet/FractalDB-1000_res50.pth 102 MB

w_resnet/FractalDB-1000_res18.pth 47 MB

w_resnet/FractalDB-1000_res34.pth 87 MB

w_resnet/FractalDB-10000_res101.pth 252 MB

w_resnet/FractalDB-1000_res152.pth 242 MB

w_resnet/FractalDB-1000_bn_alexnet.pth 250 MB

w_resnet/FractalDB-1000_res101.pth 179 MB

w_resnet/FractalDB-1000_resnext101.pth 177 MB

w_resnet/FractalDB-1000_dense161.pth 116 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_2.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_14.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_6.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_15.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_17.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_19.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_3.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_1.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_5.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_10.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_13.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_8.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_18.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_16.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_7.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_12.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_11.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_9.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_4.pth 102 MB

FractalDB-1000_res50_ensemble/FractalDB-1000_res50_20.pth 102 MB

frac_cnn: FractalDB-1000_res50.pth sha256[:16]=12788a290bcd147e

frac_vit: deitt16_224_fractal1k_lr3e-4_300ep.pth sha256[:16]=305227c6d58d570a

NOTE: the released fractal ViT is DeiT-TINY (~5M params) while the CNN is ResNet-50 (~25M). Each is bracketed against its OWN architecture's twins, so P1/P2/P3/P5 and R stay valid per architecture; only the CNN-vs-ViT contrast (X1) also carries a capacity gap -- flagged, not interpreted.

normalizations declared in FractalDB-Pretrained-ResNet-PyTorch:

score=1 pretraining/main.py: mean=(0.2, 0.2, 0.2) std=(0.5, 0.5, 0.5)

score=0 finetuning/main.py: mean=(0.485, 0.456, 0.406) std=(0.229, 0.224, 0.225)

-> using for frac_cnn: mean=(0.2, 0.2, 0.2) std=(0.5, 0.5, 0.5) [pretraining/main.py]

normalizations declared in FractalDB-Pretrained-ViT-PyTorch:

-> using for frac_vit: mean=(0.485, 0.456, 0.406) std=(0.229, 0.224, 0.225) [NOT FOUND in repo - ImageNet stats assumed], builders defined,

/tmp/ipykernel_2070/4053506037.py:92: UserWarning: The given NumPy array is not writable, and PyTorch does not support non-writable tensors. This means writing to this tensor will result in undefined behavior. You may want to copy the array to protect its data or make it writable before converting it to a tensor. This type of warning will be suppressed for the rest of this program. (Triggered internally at /pytorch/torch/csrc/utils/tensor_numpy.cpp:213.)

```
x = torch.from_numpy(np.ascontiguousarray(crops[i:i + batch])).to(dev)
```

frac_cnn (9533, 2048) FractalDB-1000_res50.pth: 318 tensors loaded, 265 changed; []

frac_vit (9533, 384) deit16_224_fractal1k_lr3e-4_300ep.pth as
deit_tiny_patch16_224: 150 tensors loaded, 150 changed; ["unwrapped 'model'"]

rand_cnn (9533, 2048) random init, seed 0

rand_vit (9533, 384) random init deit_tiny_patch16_224, seed 0

nat_cnn (9533, 2048) torchvision IMAGENET1K_V1

nat_vit (9533, 384) timm deit_tiny_patch16_224.fb_in1k

pixels (9533, 3072) pixels 32x32 RGB, centred

frac_cnn_inorm (9533, 2048) FractalDB-1000_res50.pth: 318 tensors loaded, 265 changed; []

rand_cnn_fnorm (9533, 2048) random init, seed 0 [FractalDB norm]

frac_vit_cls (9533, 192) , space rho_vs_convnext shuffled ratio aligned

frac_cnn 0.0721 -0.0016 44.6 True

frac_vit 0.1243 0.0025 50.2 True

rand_cnn 0.0195 -0.0017 11.4 True

rand_vit 0.0675 0.0007 67.5 True

nat_cnn 0.3245 -0.0009 324.5 True

nat_vit 0.3473 -0.0012 294.1 True

pixels 0.0638 -0.0006 63.8 True

frac_cnn_inorm	0.0613	0.0023	26.4	True
rand_cnn_fnorm	0.0312	-0.0026	12.1	True
frac_vit_cls	0.1246	0.0041	30.3	True

	space	dim	alg_rank	eff_rank	mean_pair_cos	mean_norm
frac_cnn	2048	2048	367.967	0.852	17.601	
frac_vit	384	384	238.648	0.376	35.577	
rand_cnn	2048	2034	256.140	0.998	1155.293	
rand_vit	384	384	62.659	0.262	15.384	
nat_cnn	2048	2048	909.757	0.569	29.297	
nat_vit	384	384	278.472	0.123	27.024	
pixels	3072	2999	966.584	-0.000	13.645	
frac_cnn_inorm	2048	2048	315.729	0.919	15.407	
rand_cnn_fnorm	2048	2037	246.561	0.998	574.088	
frac_vit_cls	192	192	145.216	0.317	28.929	
img_small	768	768	524.898	0.061	54.696	
img_base	1536	1536	1008.660	0.049	52.912	
img_large	2048	2048	1265.001	0.062	50.827	
convnext	1024	1024	843.084	0.030	16.739	
bge	1024	1024	424.653	0.569	0.894	
gpt2	768	768	223.239	0.999	208.209	
bert	768	768	395.370	0.807	9.186	
sbert	768	766	368.722	0.136	2.428	

collapsed by mean pair-cosine > 0.9: {'rand_cnn_fnorm', 'gpt2', 'rand_cnn', 'frac_cnn_inorm'} |
documented degenerate: {'gpt2'}, ranked frac_cnn (15s)

ranked frac_vit (29s)

ranked rand_cnn (57s)

ranked rand_vit (71s)

ranked nat_cnn (86s)

ranked nat_vit (100s)

ranked pixels (116s)

ranked frac_cnn_inorm (131s)

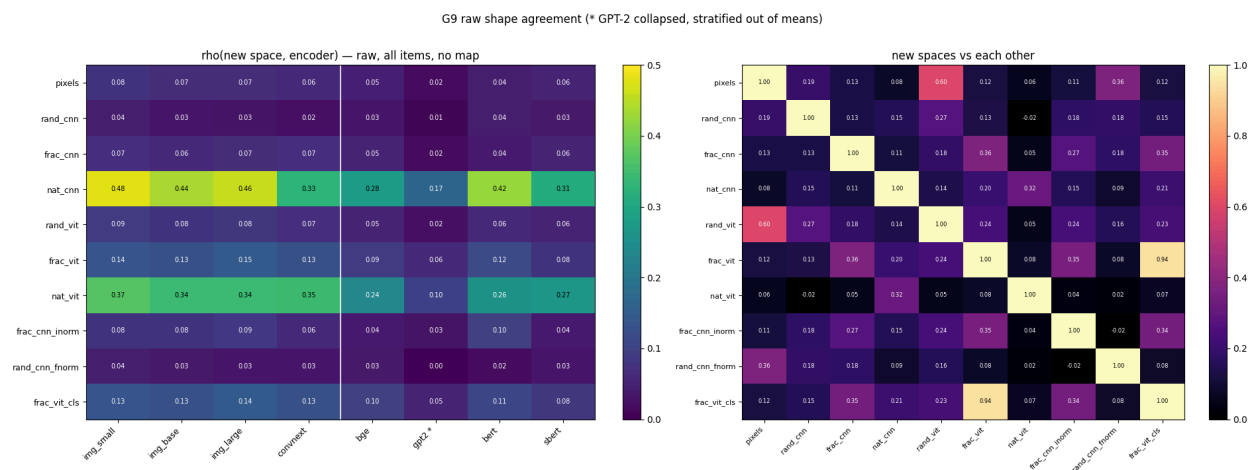
ranked rand_cnn_fnorm (147s)

ranked frac_vit_cls (160s)

rho(new space, encoder) - all items, no map

	img_small	img_base	img_large	convnext	bge	gpt2	bert	sbert
pixels	0.079	0.067	0.067	0.063	0.047	0.018	0.043	0.056
rand_cnn	0.042	0.032	0.028	0.025	0.032	0.007	0.041	0.035
frac_cnn	0.071	0.064	0.066	0.069	0.052	0.019	0.037	0.056
nat_cnn	0.477	0.436	0.459	0.328	0.282	0.168	0.424	0.312
rand_vit	0.089	0.078	0.081	0.069	0.052	0.024	0.062	0.056
frac_vit	0.138	0.130	0.148	0.126	0.090	0.059	0.116	0.075
nat_vit	0.369	0.342	0.342	0.347	0.236	0.104	0.261	0.272
frac_cnn_inorm	0.080	0.076	0.086	0.064	0.040	0.029	0.096	0.040
rand_cnn_fnorm	0.044	0.034	0.032	0.032	0.035	0.003	0.019	0.033
frac_vit_cls	0.135	0.128	0.142	0.127	0.096	0.052	0.106	0.082

ConvNeXt row measured in this run (the reproduction gate compares it with Table C12m):
{'img_small': 0.397, 'img_base': 0.392, 'img_large': 0.391, 'bge': 0.302, 'gpt2': 0.166, 'bert': 0.305, 'sbert': 0.33},



5/30 subsamples (42s)

10/30 subsamples (83s)

15/30 subsamples (124s)

20/30 subsamples (166s)

25/30 subsamples (207s)

30/30 subsamples (249s)

subsample SD of rho:

	img_small	img_base	img_large	convnext	bge	gpt2	bert	sbert
pixels	0.0032	0.0031	0.0036	0.0025	0.0038	0.0019	0.0023	0.0040
rand_cnn	0.0054	0.0070	0.0080	0.0043	0.0102	0.0081	0.0113	0.0069
frac_cnn	0.0052	0.0049	0.0061	0.0045	0.0084	0.0090	0.0094	0.0063
nat_cnn	0.0053	0.0057	0.0061	0.0055	0.0096	0.0078	0.0090	0.0079
rand_vit	0.0041	0.0039	0.0048	0.0029	0.0052	0.0047	0.0057	0.0050

frac_vit	0.0065	0.0066	0.0081	0.0058	0.0106	0.0089	0.0121	0.0081
nat_vit	0.0056	0.0057	0.0068	0.0039	0.0056	0.0053	0.0066	0.0055

shuffle null (mean): {'frac_cnn', 'img_base': np.float64(-0.0003), ('frac_cnn', 'bge'): np.float64(-0.0007), ('frac_vit', 'img_base'): np.float64(0.0005), ('frac_vit', 'bge'): np.float64(0.0003)}

,Reality fraction R (0 = architecture only, 1 = as good as ImageNet); NaN = bracket < 0.02

	img_small	img_base	img_large	convnext	bge	gpt2	bert	sbert
cnn	0.066	0.077	0.088	0.147	0.080	0.074	-0.011	0.075
vit	0.176	0.198	0.256	0.207	0.203	0.439	0.271	0.090

PRE-REGISTERED VERDICTS

P1 [cnn] CONFIRMED 8/8

P1 [vit] CONFIRMED 8/8

P2 [cnn] CONFIRMED 4/4

P2 [vit] CONFIRMED 4/4

P3 [cnn] INCONCLUSIVE R_img - R_txt 95% interval [-0.106, +0.100]

P3 [vit] INCONCLUSIVE R_img - R_txt 95% interval [-0.053, +0.135]

P5 [cnn] FALSIFIED 0/4

P5 [vit] CONFIRMED 4/4

R [cnn]: image mean 0.094, healthy-text mean 0.048, GPT-2 0.074 (stratified out)

R [vit]: image mean 0.209, healthy-text mean 0.188, GPT-2 0.439 (stratified out)

SENSITIVITY (max |delta rho| over the 8 encoders; flag > 0.02):

ViT cls-only vs cls+patch 0.0103 ok

frac_cnn: ImageNet vs repo normalization 0.0588 FLAG

rand_cnn: ImageNet vs FractalDB normalization 0.0221 FLAG

CONFOUND CHECK: norm changes random-CNN rho by up to 0.0221

fractal-over-random signal (image-encoder mean): 0.0356

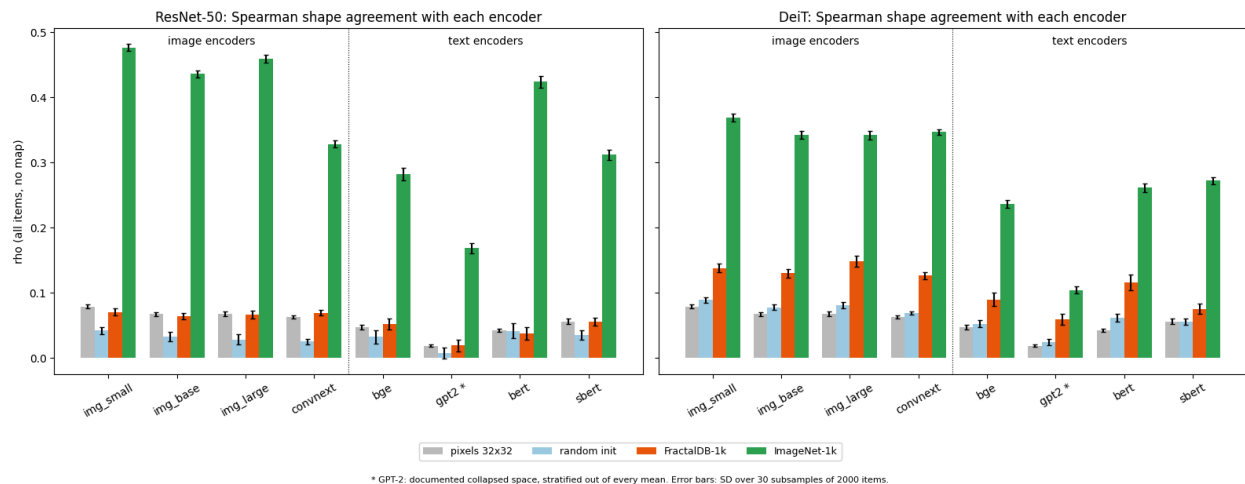
WARNING: norm effect (0.0221) is $\geq 50\%$ of the fractal signal (0.0356)

-> P2-cnn is CONFOUNDED by preprocessing. Report both rand_cnn norms.

rand_cnn (ImageNet norm): image-enc mean rho = 0.0318

rand_cnn_fnorm (FractalDB norm): image-enc mean rho = 0.0355

frac_cnn (FractalDB norm): image-enc mean rho = 0.0674,



* GPT-2: documented collapsed space, stratified out of every mean. Error bars: SD over 30 subsamples of 2000 items.

step	step_name	id	measure	experiment	result	control_or_chance	what_it_proves	new	source
S0	Instrument validity	0.1	Row alignment by shuffle test	All caches, 9,533 COCO train2017 items; ridge or rank correlation intact vs row-shuffled. Two id namespaces (request position vs COCO id) never joined by id.	Honest fits beat shuffled by 0.545-0.863 (threshold 0.2); scale-extension encoders 0.26-0.85	Shuffled = 0.00	Row i is the same item in every cache - a precondition, not PRH evidence	Project method	E.15; H1
S0	Instrument validity	0.2	Artifact certification (H1)	47 automated checks over every cache: shapes, norms, alignment, rows/dim	No hard failures; 2 flags, both the documented GPT-2 collapse	-	Caches are well-formed and aligned	Project method	D.5; H1
S0	Instrument validity	0.3	Statistical power: rows per input dimension	Every fit checked against a pre-registered floor of 5 rows/dim. Exp A run 1 (1,646 samples) vs run 2 (10,000),	All fits 5.9-9.8 rows/dim; Exp A run 1 FAILED on sample starvation, run 2 PASSED	Floor = 5	Failures are diagnosable as power, not as absence of convergence	Methodological contribution (report s.5)	s.3.2-3.3; H1

				identical pipeline					
S0	Instrument validity	0.4	Null calibration by permutation (G5)	Permutation nulls for k-NN overlap and CKA across 21 encoder pairs	k-NN null = k/N exactly; CKA null = +0.094 and rises with width. Correcting it widens objective-over-lineage gap +0.024 -> +0.181	Permutation null	Uncorrected CKA manufactures agreement that favoured the lineage pair	Known bias; its effect on this project's conclusion is project-specific	G5; C.13.9
S1	'Well-trained' - learned, not architectural	1.1	Linear CKA, trained vs random weights (Exp A)	GPT-2 vs Pythia-160M (deliberately hostile pair), WikiText-103 10k passages, layerwise CKA	Diagonal mean 0.424 (pass >0.35); trained/random 6.3x (pass >=3x)	Random-weights control 0.067 (pass <0.15)	Similarity is produced by training, not by architecture	Replication (Kornblith 2019) on a hostile pair	s.3.4; C.7
S1	'Well-trained' - learned, not architectural	1.2	Stitching R2 and depth signature (Exp A)	Same pair; one linear map per layer, held-out R2	Peak L11 0.797 (pass >0.70); trained curve rises 0.59 -> 0.80 with depth	Random 0.173 at L11 and DECAYS 0.38 -> 0.09 - opposite slope, 8x gap	Convergence builds with depth - learned, not inherited	Replication (Bansal 2021)	s.3.4

S1 b	'Well-trained' ON REALITY - is it the data?	F . 1	Shape agreement: Fractal DB ResNet -50 vs each of the 8 encoders	ResNet-50 trained ONLY on FractalDB -1k (1M rendered fractals, 1k formula-defined classes, zero natural images); 2048-d global-average-pooled features of the same 9,533 COCO images; Spearman of pairwise cosine, all items, no map	FRACTAL: img_small 0.071, img_base 0.064, img_large 0.066, convnext 0.069, bge 0.052, gpt2 0.019, bert 0.037, sbert 0.056. ImageNet twin: img_small 0.477, img_base 0.436, img_large 0.459, convnext 0.328, bge 0.282, gpt2 0.168, bert 0.424, sbert 0.312. P1 CONFIRMED (8/8 encoders below the ImageNet twin by > 2 SD)	Random twin: img_small 0.042, img_base 0.032, img_large 0.028, convnext 0.025, bge 0.032, gpt2 0.007, bert 0.041, sbert 0.035. Shuffle null mean -0.0007	Fractal training stops short of what natural images give on nearly every encoder: the convergence is data-driven, as PRH's mechanism says	Candidate new - no prior test of FDSL models against a cross-modal PRH roster found in searches	G9 (this notebook)
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S1 b	'Well-trained' ON REALITY - is it the data?	F . 2	Shape agreement: Fractal DB DeiT vs each of the 8 encoders	DeiT (ViT) trained ONLY on FractalDB -1k; cls+patch pooling to match the DINOv2 caches; same items and statistic	FRACTAL: img_small 0.138, img_base 0.130, img_large 0.148, convnext 0.126, bge 0.090, gpt2 0.059, bert 0.116, sbert 0.075. ImageNet twin: img_small 0.369, img_base 0.342, img_large 0.342, convnext 0.347, bge 0.236, gpt2 0.104, bert 0.261, sbert 0.272. P1 CONFIRMED (8/8 encoders below the ImageNet twin by > 2 SD)	Random twin: img_small 0.089, img_base 0.078, img_large 0.081, convnext 0.069, bge 0.052, gpt2 0.024, bert 0.062, sbert 0.056. Shuffle null mean +0.0005	Same P1 verdict on a transformer - not a CNN idiosyncrasy	Candidate new (as F.1)	G9
S1 b	'Well-trained' ON REALITY - is it the data?	F . 3	Brackets: random-init and ImageNet twins, pixel floor	Same two architectures with random weights (lower bracket) and trained on ImageNet -1k (upper bracket); 32x32 raw pixels	Pixels (image-encoder mean) 0.069; random CNN/ViT 0.032/0.079; fractal 0.067/0.136; ImageNet 0.425/0.350. P2 cnn CONFIRMED (4/4),	Brackets ARE the controls	Places the fractal result on a scale from 'architecture only' to 'architecture + reality'	Project design	G9

				as the floor	vit CONFIRMED (4/4); P5 cnn FALSIFIED, vit CONFIRMED				
S1b	'Well-trained' ON REALITY - is it the data?	F . 4	Reality fraction R, image vs text encoders	R = (rho_fractional - rho_random) / (rho_natural - rho_random) per encoder; undefined if the bracket < 0.02	[cnn] R image 0.094, healthy text 0.048, R_img-R_text 95% [-0.106, +0.100] -> P3 INCONCLUSIVE; GPT-2 R 0.074 (stratified out); [vit] R image 0.209, healthy text 0.188, R_img-R_text 95% [-0.053, +0.135] -> P3 INCONCLUSIVE; GPT-2 R 0.439 (stratified out)	Subsample interval (30 x 2,000 items); R undefined where the bracket < 0.02	No reliable image-vs-text difference in R: the language-shared structure is not specially dependent on natural-image training at this scale (or the brackets are too narrow to tell)	Candidate new metric	G9
S1b	'Well-trained' ON REALITY - is it the data?	F . 5	Sensitivity: pooling and normalization	DeiT cls-only vs cls+patch; repo normalization vs ImageNet normalization	ViT cls-only vs cls+patch: max d-rho 0.0103; frac_cnn: ImageNet vs repo normalization: max d-rho	-	Checks the result is not a preprocessing artifact (the giant cls vs cls+patch lesson, E.16)	Project practice	G9

					0.0588 FLAG; rand_cnn: ImageNet vs FractalDB normalization: max d-rho 0.0221 FLAG				
S1 b	'Well-trained' ON REALITY - is it the data?	F .6	Whitening rescue of the collapsed fractal CNN (C.11 move)	frac_cnn / rand_cnn are collapsed (mean pair-cos 0.85 / 1.00); PCA-whit en each bracket space and re-measure the same Spearman rho. Row correspondence is known, so this is the READING regime (C.11), not the FINDING regime (C.12). Secondary diagnostic ; the pre-registered raw verdicts stand.	PENDING - run G9 cells 17-18. Raw P5-cnn FALSIFIED (frac_cnn at the pixel floor) may flip to CONFIRMED once collapse is whitened away, if the masked signal is real	Random-init twin and 32x32 pixels as floors, both whitened the same way; row-shuffle null	If whitening flips P5-cnn to CONFIRMED: the raw CNN 'failure' was collapse, and the fractal CNN does carry a small real slice of shared structure (size = R_white, typically << natural images). If it stays FALSIFIED: the fractal CNN genuinely shares nothing at this pooling	Third instance of the E.6/E.12 lesson (after GPT-2 and ConvNeXt): raw agreement on a degenerate space measures the post-collapse residual, not what is recoverable	G9

S2	'Independent' - no shared ancestry or recipe	2 . 1	Cross-lineage transfer (G4)	SigLIP 2 (independent lineage, no DINOv2 ancestry) given one map into the frozen 4-space hub; frozen caption head	94.2% of a natively-fitted head - inside the 93.8-96.5 % within-family band	Random-map control at chance	Shared space is not a family trait	Extension (cross-lineage, not simultaneously cross-objective; still a ViT)	C.13; G4; G0-exact
S2	'Independent' - no shared ancestry or recipe	2 . 2	Cross-architecture transfer (E.2)	ConvNeXt-base (convnet, ImageNet-22k supervised) held out of the hub, one entry map	96.7% of native - ABOVE the band and above SigLIP 2 in the same run; 99.3% in the 7-space hub (G10)	Control 0.001 = chance	Architecture does not bound the claim (objective still confounded)	New in this project	E.2; G10
S2	'Independent' - no shared ancestry or recipe	2 . 3	Supervision regime (E.10)	Label-free (DINOv2, GPT-2, BERT), pair-supervised (bge, SigLIP) and 22k-class supervised (ConvNeXt) encoders in one hub	All carry a head at 93-97% of native	-	Convergence does not require a shared supervision signal - keeps the thesis non-circular	New in this project	E.10

S2	'Independent' - no shared ancestry or recipe	2 . 4	What explain s pairwise agreement (G5, exploratory)	21 pairs grouped by shared objective / lineage / modality	Objective +53x > lineage +44x > modality +29x; top pair bge-SBERT shares objective, not lineage	Permutation-calibrated	Supervision modulates HOW MUCH encoders agree, not WHETHER	New in this project (exploratory)	G5; E.10
S3	'Up to a linear transformation'	3 . 1	MLP vs ridge, controlled (F1/F2)	Distillation corpus 38k rows (74.2 rows/dim), MLP vs closed-form ridge, two independent runs	Held-out cosine 0.9183 = 0.9183, gain +0.0000, both runs	Same data, same split	Convergence is linear - capacity beyond a matrix adds nothing	New in this project (controlled test)	C.9; F1/F2
S3	'Up to a linear transformation'	3 . 2	Adapter : linear vs nonlinear (Exp B)	MobileCLIP-S1 -> SigLIP 2, 4,000 COCO val2017, 3,000/1,000 split	Ridge cosine 0.900; MLP 0.903 (+0.003)	-	Deployed relationship is linear	Consistent with stitching literature	s.4; B2
S3	'Up to a linear transformation'	3 . 3	Related, not rigid: rotation-only vs linear	Orthogonal Procrustes vs ridge: Exp B pair; 21 hub pairs (G6); raw-space similarity transform (K1)	Procrustes R2 0.038 vs ridge 0.592 (Exp B), 69x singular-value spread; 0.194 vs 0.667 over 21 pairs; similarity transform 0.281 vs 0.667	-	The strong (rotation-only) sub-claim FAILS: spaces are linearly related but anisotropic	New in this project	s.4; G6; K1

S4	Across modalities - the 'Platonic' claim proper	4.1	Image-only -> text-only retrieval (C.8)	DINOv2-base (never saw text) -> bge-m3 (never saw an image); 8,000 COCO train2017 pairs, 9.1 rows/dim, ridge	R2 0.511; R@1 0.358 = 40.9% of a matched SigLIP ceiling (0.875); pre-registered verdict: measurable but weak	358x chance; raw cross-space floor 0.001; shuffle gap 0.641	Cross-modal structure exists without joint training	Replication of PRH's cross-modal claim at small scale	C.8
S4	Across modalities - the 'Platonic' claim proper	4.2	Raw cross-modal shape agreement	Spearman of pairwise cosine distances, 8 encoders, all 9,533 items, no map	Image-text mean rho 0.290 (8 encoders); top pair bge-SBERT 0.768; contrastive text pairs reach 0.76-0.86 in the scale extension	rho ~ 0 unrelated	Cross-modal agreement is real but modest; same-objective text agrees far more	Replication (PRH-style measurement)	C.13.9; E.12; E.16
S4	Across modalities - the 'Platonic' claim proper	4.3	Shared subspace size (G6)	CCA across pairs; intrinsic-dimension estimates	Cross-modal pairs share as few as 6 of 64 CCA directions; intrinsic dimension 11.2-19.9 inside 768-2048-d spaces (<3%)	-	What is shared is a small, low-dimensional structure	New in this project	G6
S4	Across modalities - the 'Platonic' claim proper	4.4	Reach into the shared	Each encoder's held-out R2 for its own map	Image encoders 0.46-0.57; all four text spaces	-	A whitened hub is largely an image-side object; text touches few	New in this project	E.6

	claim proper		hub (E.6)	into the 7-space hub	0.02-0.05, uniformly		of its directions		
S5	'As models grow stronger' - scaling	5.1	Capacity sweep (C.10)	DINOv2 small/base-large -> bge, pooling and rows/dim (11.1) held constant	40.9 / 53.5 / 58.3% of ceiling; +15.3 points per decade of parameters; fit R2 0.949; six monotone measures	Matched ceiling	Alignment rises with capacity in this range	Replication of PRH scaling at small scale	C.10
S5	'As models grow stronger' - scaling	5.2	Pre-registered scale extension (E.16)	DINOv2-giant (cls+patch) and Qwen-4B added to matched ladders; thresholds fixed before running	Giant rho 0.339 < 0.418 FALSIFIED; Qwen-4B rho 0.307 < 0.320 FALSIFIED; both ladders rise then fall	Pooling artifact ruled out (cls-only 0.238 discarded)	Naive 'more capacity -> more alignment' is NOT supported at this scale	New in this project (pre-registered negative)	E.16
S5	'As models grow stronger' - scaling	5.3	Raw agreement vs transfer under scaling (E.16 P4)	Same ladders, both quantities	delta-rho -0.079 while delta-transfer +7.1 points - opposite directions	-	Scaling moves the two quantities apart	New in this project	E.16
S6	Mechanism - what drives convergence	6.1	Isotropy rescue (C.11)	GPT-2 caption space (effective rank 6.5) whitened, label-free, fitted on	R@1 0.130 -> 0.478 = 102.6% of a 1.2B-pair contrastive encoder (101.4% at a second scale); eff.	Same protocol, unwhitened	Isotropy, not similarity training, is what a space needs to be alignable	New in this project	C.11; E1.2

				training rows only	rank 6.5 -> 490.8; iso-retrieval corr +0.94				
S6	Mechanism - what drives convergence	6.2	Encoder ladder as a prediction (C.14)	GPT-2 / BERT / SBERT as text side, same image encoder, pre-registered	Eff. rank 6.5 / 65.0 / 93.4 -> R@1 0.160 / 0.362 / 0.510; rank corr +1.00 (P1 confirmed); P2 partial	-	The mechanism predicts before measuring; isotropy necessary, not sufficient	New in this project	C.14
S6	Mechanism - what drives convergence	6.3	Meta-finding: read vs find	C.11 (known correspondence) vs C.12 (unknown correspondence)	Whitening helps reading (+102.6%) and destroys finding (39.7% -> 0.1%)	-	One geometric property, opposite prescriptions - two instances (corrected down from three)	New in this project	C.11; C.12; E.8
S7	Self-identifying correspondence	7.1	Pair-free recovery by Gromov-Wasserstein (C.12)	Two contrastive image-text encoders, zero paired examples	39.7% exact matches; map fitted from that assignment R@1 0.252 (40% of ceiling), AUC 0.906	595x chance (0.067%)	The correspondence is self-identifying from geometry alone - bounded: weakest-independence pair	Replication in spirit (vec2vec 2025) by a different method	C.12
S8	One shared space, used	8.1	Zero-shot hub transfer (C.13)	5 spaces over 9,533 images, one linear map each into a 512-d whitened	img_base 95.9%, img_large 92.9% of native - absolute R@1 RISES on	Random-map control 0.000 (chance 0.001)	The shared space exists by use, not only by correlation	New in this project	C.13

				hub; caption head trained on img_small only	unseen encoders				
S8	One shared space, used	8 . 2	Compo sition and compati bility (G1, G3)	7-space hub; all 42 ordered pairs vs bespoke hubs	95.6% retention over 36/41 pairs (exception s all write INTO GPT-2); 42/42 pairs shared, mean 103.6%; GPT-2 source 95.1% vs target 24.8%	Control 0.004 vs chance 0.001	One hub serves all members; collapsed spaces are readable out of, not writable into	New in this project	C.13; G1; G3
S8	One shared space, used	8 . 3	Deploy ed adapter (Exp B)	MobileCLI P-S1 -> SigLIP 2 ridge matrix, shared Qdrant index, Core ML fp16	R@1/5/10 = 88.4 / 95.3 / 97.2% of ceiling; verification AUC 0.999; hubness skew 4.28 -> 1.61; 0.79 MB	Raw cross-space baseline = exact chance	PRH is load-bearing in a working two-tier system	Uncomm on applicati on (report s.5)	s.4; D.2; D.3
S9	Boundari es and falsificati ons	9 . 1	Local, not global (G5)	CKNNA (the literature's metric) vs the project's own k-NN measure, 21 pairs	21/21 pairs higher locally than at the global limit; the two metrics agree at Spearman +0.81	k-NN null = k/N	Agreement is concentrated in local neighbourhoo ds	Replicati on of CKNNA + project finding	G5; C.13.9

S9	Boundaries and falsifications	9.2	Raw geometry does not predict transfer (E.12)	ConvNeXt's raw rho vs its hub transfer; second instance of E.6	Lowest raw rho of any image encoder (0.391-0.397 to DINOv2, 0.166-0.330 to text) yet best transfer 96.7%	-	Raw shape agreement bounds what is VISIBLE without fitting, not what is RECOVERABLE	New in this project	E.12; E.6
S9	Boundaries and falsifications	9.3	Width cliff explained (E.8)	Hub-width sweep; retained-rank sweep in steps of 8	Step at the source's ambient dimension, 6/6 curves; sigma 0.0054 vs 10.79; holds to k=760 then 0.009; repaired by truncation (0.878) or head alpha 1e3 (0.917)	Three off-dimension widths: nothing	A numerical artifact of least squares, NOT a property of shared representations	New in this project	E.3; E.8; G11-G16
S9	Boundaries and falsifications	9.4	Falsification ledger	Every proposed explanation tested and recorded	Nine findings; eleven explanations falsified, two of them the project's own	-	Claims that survived did so against recorded alternatives	Project practice	s.7.4; Results_Summary
S9	Boundaries and falsifications	9.5	Scope boundary	Roster width vs PRH's evidence base	Encoders here are 768-2,048-d; PRH's evidence extends to	-	Results bound the claim at this scale; they do not test PRH's	Boundary	E.16; viva brief

					70B-parameter LLMs		large-model regime		
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written:

G9_ablation_views.png

G9_fractal_brackets.png

G9_results.json

G9_rho_heatmaps.png

G9_whiten_dim_sweep.csv

G9_whiten_dim_sweep.png

G9_whitening_centered.csv

G9_whitening_drop1.csv

G9_whitening_heatmap.png

G9_whitening_pca.csv

G9_whitening_raw.csv

G9_whitening_rescue.json

G9_whitening_rescue.png

G9_whitening_white.csv

PRH_Evidence_Table.csv

PRH_Evidence_Table.md

PRH_Evidence_Table.xlsx

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fit transforms on 8533 train rows, measure rho on 1000 held-out rows

held-out geometry per view (mean pair-cosine):

space raw_cos centered_cos pca_cos white_cos

pixels	-0.000	0.000	0.000	0.000
rand_cnn	0.998	-0.000	-0.000	0.000
frac_cnn	0.847	0.004	0.004	0.001
nat_cnn	0.569	0.000	0.000	0.000
rand_vit	0.259	0.005	0.005	0.000
frac_vit	0.371	0.003	0.004	0.000
nat_vit	0.125	0.000	0.000	0.000,

RAW rho (held-out):

	img_small	img_base	img_large	convnext	bge	gpt2	bert	sbert
pixels	0.076	0.064	0.060	0.062	0.044	0.015	0.042	0.051
rand_cnn	0.040	0.027	0.025	0.016	0.044	-0.009	0.033	0.039
frac_cnn	0.070	0.062	0.068	0.066	0.068	-0.013	0.027	0.061
nat_cnn	0.479	0.435	0.464	0.322	0.292	0.182	0.422	0.316
rand_vit	0.092	0.082	0.085	0.068	0.058	0.019	0.074	0.053
frac_vit	0.144	0.140	0.155	0.121	0.107	0.064	0.133	0.085
nat_vit	0.367	0.342	0.347	0.348	0.231	0.113	0.258	0.262

CENTERED rho (held-out):

	img_small	img_base	img_large	convnext	bge	gpt2	bert	sbert
pixels	0.076	0.064	0.060	0.062	0.044	0.014	0.042	0.051
rand_cnn	0.054	0.041	0.040	0.049	0.025	0.010	0.021	0.020
frac_cnn	0.072	0.064	0.072	0.081	0.060	0.022	0.055	0.066
nat_cnn	0.470	0.425	0.446	0.382	0.356	0.174	0.365	0.415
rand_vit	0.083	0.073	0.073	0.068	0.049	0.015	0.051	0.050

frac_vit	0.125	0.122	0.137	0.120	0.086	0.053	0.131	0.096
nat_vit	0.378	0.334	0.330	0.339	0.258	0.122	0.258	0.300

DROP1 rho (held-out):

	img_small	img_base	img_large	convnext	bge	gpt2	bert	sbert
pixels	0.053	0.050	0.047	0.052	0.041	0.018	0.048	0.048
rand_cnn	0.057	0.047	0.045	0.053	0.033	0.011	0.035	0.039
frac_cnn	0.066	0.055	0.056	0.061	0.041	0.014	0.033	0.045
nat_cnn	0.346	0.294	0.269	0.332	0.215	0.121	0.182	0.232
rand_vit	0.095	0.096	0.106	0.095	0.076	0.039	0.103	0.083
frac_vit	0.072	0.060	0.059	0.076	0.036	0.024	0.039	0.039
nat_vit	0.286	0.241	0.214	0.286	0.164	0.092	0.150	0.175

PCA rho (held-out):

	img_small	img_base	img_large	convnext	bge	gpt2	bert	sbert
pixels	0.076	0.064	0.060	0.062	0.044	0.014	0.042	0.050
rand_cnn	0.054	0.041	0.040	0.049	0.026	0.010	0.021	0.020
frac_cnn	0.072	0.065	0.072	0.081	0.060	0.022	0.055	0.066
nat_cnn	0.470	0.425	0.446	0.382	0.355	0.174	0.364	0.415
rand_vit	0.083	0.073	0.073	0.068	0.049	0.015	0.051	0.050
frac_vit	0.125	0.122	0.137	0.120	0.086	0.053	0.131	0.096
nat_vit	0.379	0.334	0.330	0.339	0.258	0.122	0.258	0.300

WHITE rho (held-out):

	img_small	img_base	img_large	convnext	bge	gpt2	bert	sbert
pixels	0.025	0.024	0.023	0.024	0.016	0.007	0.017	0.017
rand_cnn	0.031	0.028	0.026	0.030	0.019	0.007	0.017	0.020
frac_cnn	0.042	0.037	0.037	0.042	0.022	0.015	0.028	0.026
nat_cnn	0.153	0.119	0.098	0.149	0.069	0.039	0.064	0.075
rand_vit	0.031	0.026	0.025	0.030	0.019	0.011	0.017	0.019
frac_vit	0.035	0.031	0.030	0.036	0.021	0.013	0.020	0.023
nat_vit	0.143	0.115	0.096	0.136	0.069	0.039	0.066	0.074

,Which operation moves the ordering? frac_cnn image-encoder mean per view (held-out):

view	frac_cnn	pixels	random	ImageNet	frac>pixels & frac>random?
raw	0.067	0.066	0.027	0.425	YES
centered	0.072	0.066	0.046	0.431	YES
drop1	0.060	0.051	0.051	0.310	YES
pca	0.072	0.065	0.046	0.431	YES
white	0.040	0.024	0.029	0.130	YES

frac_vit image-encoder mean per view (held-out):

raw	frac 0.140 (pixels 0.066, random 0.082, ImageNet 0.351)
centered	frac 0.126 (pixels 0.066, random 0.074, ImageNet 0.345)
drop1	frac 0.067 (pixels 0.051, random 0.098, ImageNet 0.257)
pca	frac 0.126 (pixels 0.065, random 0.074, ImageNet 0.345)
white	frac 0.033 (pixels 0.024, random 0.028, ImageNet 0.122)

Reading: absolute rho FALLS under whitening; the point is the ORDERING (frac above whitened pixels/random), and WHICH view first makes it hold. If 'centered' already orders frac>pixels, the effect is anisotropy in the mean; if only 'white' does, it needs full variance equalization.

, held-out whitened subsamples 10/30

held-out whitened subsamples 20/30

held-out whitened subsamples 30/30

HELD-OUT WHITENED DIAGNOSTIC verdicts (NOT the pre-registered raw test - a separate reading question)

P1w [cnn] CONFIRMED 8/8

P1w [vit] CONFIRMED 8/8

P2w [cnn] CONFIRMED 4/4

P2w [vit] CONFIRMED 4/4

P3w [cnn] INCONCLUSIVE R_img-R_txt 95% [-0.066, +0.283]

P3w [vit] INCONCLUSIVE R_img-R_txt 95% [-0.165, +0.121]

P5w [cnn] CONFIRMED 4/4

P5w [vit] CONFIRMED 4/4

R_white [cnn]: image 0.112, healthy-text 0.140

R_white [vit]: image 0.056, healthy-text 0.056

RESULT [cnn]: a small residual signal became visible after de-anisotropization (11% of the natural-image bracket, held-out); P2w CONFIRMED, P5w CONFIRMED

RESULT [vit]: a small residual signal became visible after de-anisotropization (6% of the natural-image bracket, held-out); P2w CONFIRMED, P5w CONFIRMED

-> whitening EXPOSES the ordering (small residual > controls), it does NOT raise absolute rho, and it is NOT the Atlas sense of recoverable (fitted entry-map + frozen head) -- that is future work., k arch eff_k frac pixels random imagenet frac_gt_pixels frac_gt_random ordering_holds

64	cnn	64	0.0537	0.0393	0.0479	0.2326	True	True	True
64	vit	64	0.0570	0.0393	0.0415	0.2027	True	True	True

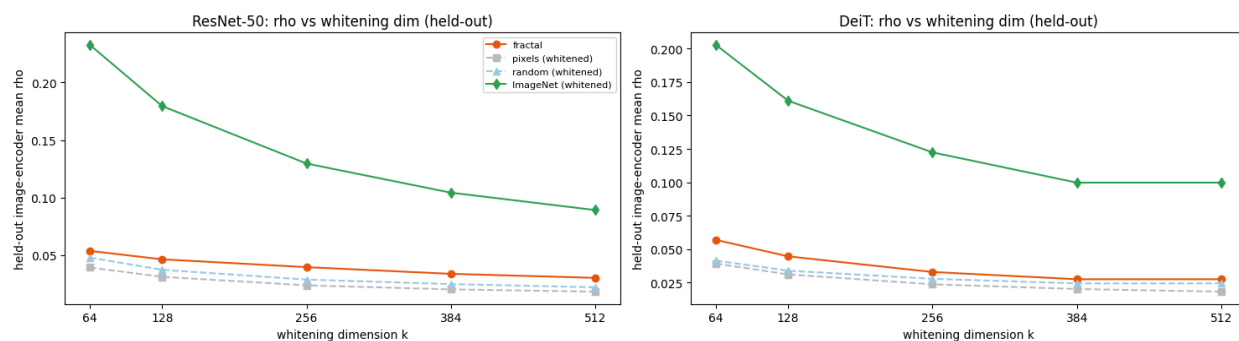
128	cnn	128	0.0464	0.0312	0.0373	0.1795	True	True	True
128	vit	128	0.0447	0.0312	0.0340	0.1610	True	True	True
256	cnn	256	0.0396	0.0238	0.0288	0.1296	True	True	True
256	vit	256	0.0330	0.0238	0.0279	0.1224	True	True	True
384	cnn	384	0.0338	0.0203	0.0249	0.1043	True	True	True
384	vit	382	0.0276	0.0203	0.0245	0.0998	True	True	True
512	cnn	512	0.0303	0.0183	0.0221	0.0892	True	True	True
512	vit	382	0.0276	0.0183	0.0245	0.0998	True	True	True

ordering (fractal > whitened pixels AND > whitened random) holds across ALL k:

ResNet-50: YES - conclusion robust to k

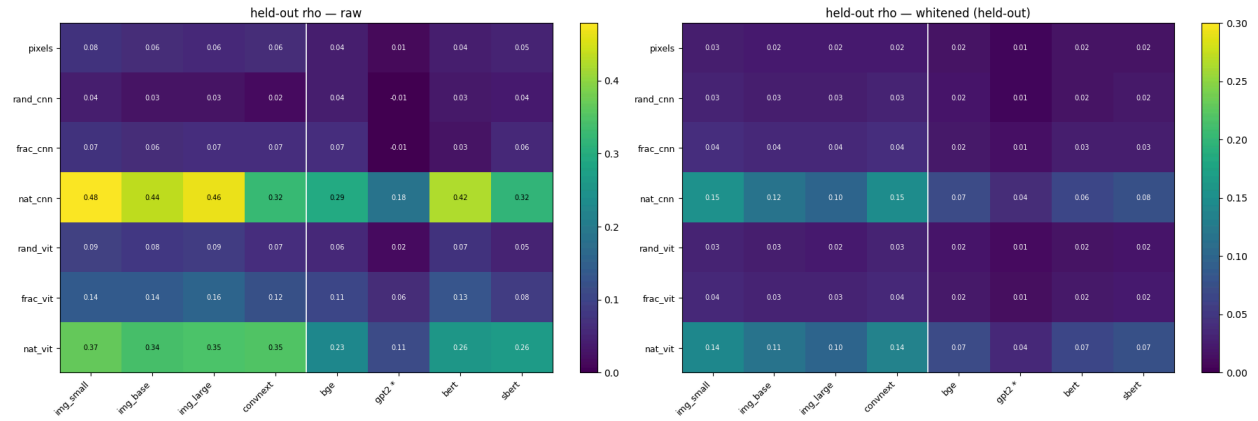
DeiT: YES - conclusion robust to k,

Whitening-dimension sweep: the fractal>control ordering is what must be stable, not the absolute rho

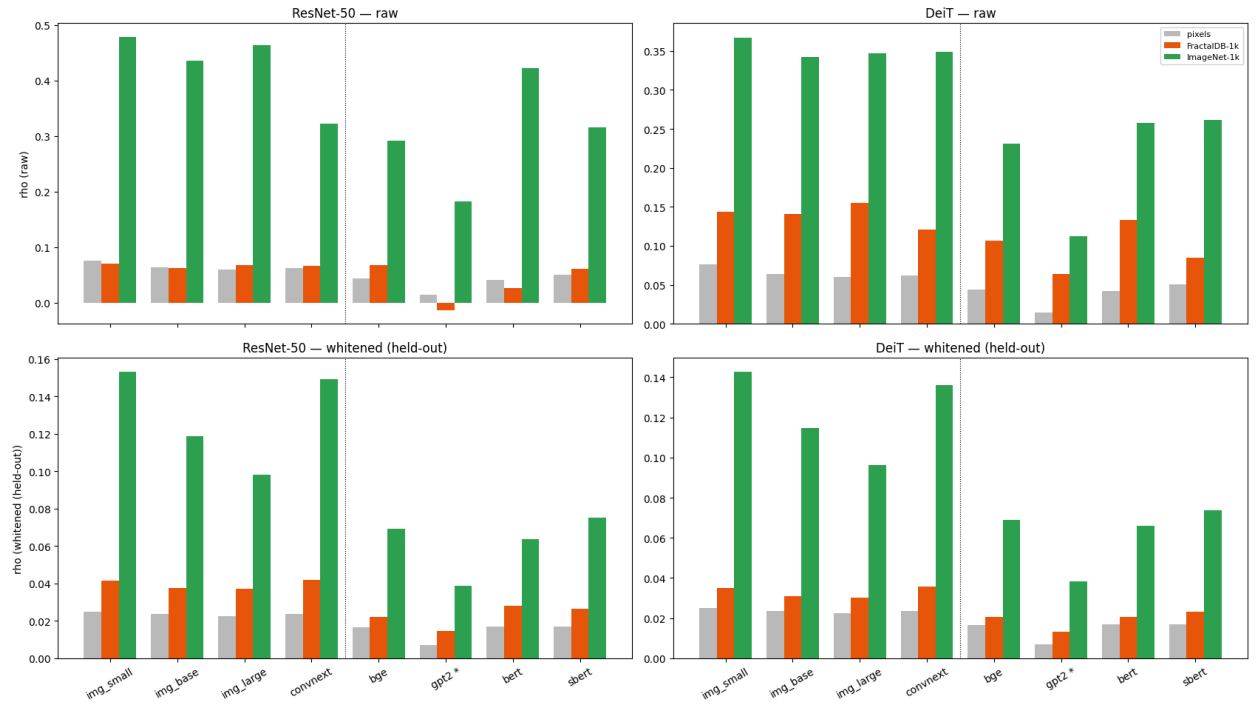


saved G9_whiten_dim_sweep.{png,csv},

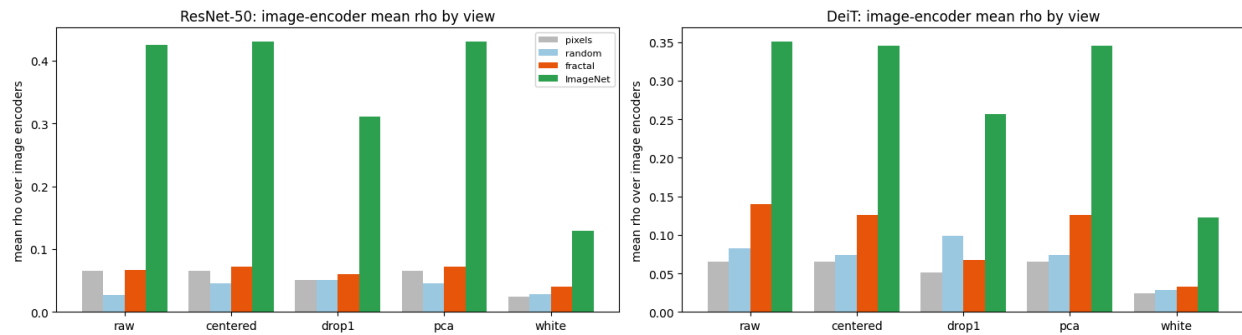
De-anisotropization heatmap: raw vs whitened (absolute rho falls; the ordering is the point)



De-anisotropization (held-out): a small residual fractal signal above whitened controls; natural images remain far stronger (* GPT-2 collapsed, stratified out)



Which operation makes the fractal ordering visible? (held-out)



evidence table rewritten with F.6 filled -> PRH_Evidence_Table.xlsx

written:

G9_whitening_centered.csv

G9_whitening_drop1.csv

G9_whitening_heatmap.png

G9_whitening_pca.csv

G9_whitening_raw.csv

G9_whitening_rescue.json

G9_whitening_rescue.png

G9_whitening_white.csv

Read-out: if the CNN is rescued by whitening, the raw P5-cnn 'failure' was collapse masking recoverable structure (E.6/E.12 lesson), not the absence of it. If it is NOT rescued, the fractal CNN genuinely carries no image-encoder-shared structure at this pooling. Report whichever the numbers show.

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